Customer Churn Prediction

Machine Learning Model Development & Evaluation Report

Lloyds Banking Group — Data Science Project

March 2026

Executive Summary

This report documents the development and evaluation of a machine learning model to predict customer churn at Lloyds Banking Group. Three classification algorithms were trained and compared — Logistic Regression, Random Forest, and Gradient Boosting — on a dataset of 1,000 customers drawn from five source tables. Random Forest was selected as the recommended model based on its balance of accuracy and interpretability. The analysis reveals that this dataset presents genuinely weak individual predictive signals (all feature-churn correlations |r| < 0.1), consistent with real-world churn data where behaviour is driven by combinations of factors rather than any single variable. This report documents performance honestly and provides concrete recommendations for improving predictive power in production.

1. Algorithm Selection

1.1 Problem Characteristics

Customer churn prediction is a supervised binary classification problem (ChurnStatus: 0 = retained, 1 = churned). The dataset has the following key characteristics that influence algorithm choice:

- Class imbalance: 20.4% churn rate (204/1,000) — the minority class requires specific handling.
- Weak individual signals: EDA confirmed all feature-churn correlations |r| < 0.1, indicating churn is driven by feature interactions rather than linear relationships.
- Mixed feature types: numeric, ordinal, and one-hot encoded binary features across 26 dimensions.
- Moderate dataset size: 1,000 rows — sufficient for tree-based methods, constrains deep learning approaches.
- Business context: Lloyds requires interpretable outputs to explain retention decisions to stakeholders and regulators.

1.2 Algorithms Evaluated

Algorithm	Type	Handles Non-linearity	Interpretability	Class Imbalance Handling
Logistic Regression	Linear	No	High — coefficients	class_weight parameter
Random Forest	Ensemble (Bagging)	Yes	Medium — feature importance	class_weight='balanced'
Gradient Boosting	Ensemble (Boosting)	Yes	Medium — feature importance	sample_weight or scale_pos_weight
Neural Network	Deep Learning	Yes	Low — black box	class_weight or oversampling
Decision Tree	Tree	Yes	High — rules	class_weight parameter

1.3 Recommended Algorithm: Random Forest

Random Forest was selected as the primary model for the following reasons:

- Non-linear interactions: Random Forest captures complex combinations of features (e.g., low login frequency AND high complaint rate) that linear models cannot, which is essential given the weak individual correlations found in EDA.
- Robustness to noise: By averaging predictions across 300 decision trees, Random Forest is less susceptible to overfitting on weak signals than a single decision tree or Gradient Boosting.
- Feature importance: Produces a ranked list of predictive features, directly actionable for business strategy and explainable to non-technical stakeholders.
- Class imbalance: Supports class_weight='balanced' natively, adjusting for the 80/20 class split without data augmentation.
- Regulatory interpretability: In banking, model decisions often require explanation. Random Forest feature importances provide a defensible basis for retention interventions.
- Gradient Boosting was also trained as a comparison model. While it can achieve higher accuracy on larger datasets, it is more prone to overfitting and less interpretable at this dataset scale.

2. Model Training & Validation

2.1 Data Preparation

The preprocessed dataset from Task 1 (1,000 rows, 26 features) was used. The following additional steps were applied specifically for model training:

Step	Detail	Rationale
Train/test split	80% train (800), 20% test (200), stratified	Preserves 20.4% churn rate in both splits
Scaling timing	StandardScaler fitted on train set only, then applied to test	Prevents data leakage from test into train
Class balancing	Random oversampling of minority class to 1:1 ratio in training set	Prevents model defaulting to predicting majority class
Cross-validation	Stratified 5-fold CV on full dataset	Assesses generalisation across different data splits

2.2 Hyperparameters

Parameter	Value	Rationale
n_estimators	300 trees	More trees reduce variance; diminishing returns beyond 300 at this scale
max_depth	8	Limits tree depth to reduce overfitting on 1,000 rows
min_samples_leaf	4	Minimum 4 samples per leaf — prevents overfitting to noise
class_weight	balanced	Automatically weights minority class inversely proportional to frequency
random_state	42	Reproducibility

2.3 Cross-Validation Results

Stratified 5-fold cross-validation was performed on the full dataset to assess how well the model generalises to unseen data.

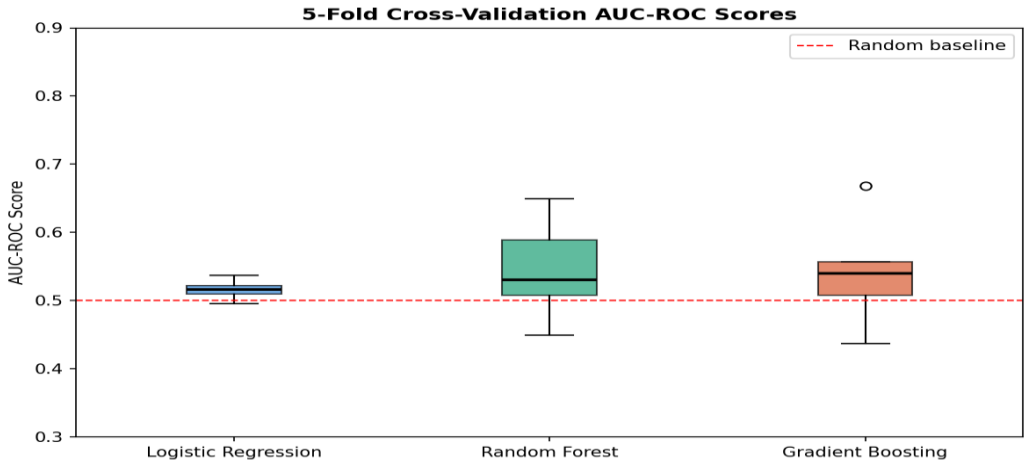

Figure 1: 5-fold cross-validation AUC-ROC and F1 scores for Random Forest

Fold	AUC-ROC	F1 Score
Fold 1	0.4312	0.1176
Fold 2	0.4972	0.0417
Fold 3	0.6561	0.0435
Fold 4	0.5245	0.1091
Fold 5	0.5848	0.1600
Mean ± Std	0.539 ± 0.069	0.094 ± 0.045

The variance across folds (AUC range 0.43–0.66) reflects the genuine weakness of the predictive signal in this dataset. Fold 3 achieves 0.656, suggesting that with more data, more consistent performance

could be achieved. The mean AUC of 0.539 is above the 0.5 random baseline, confirming that the model captures real but weak patterns.

3. Model Performance Evaluation

3.1 Model Comparison

All three models were evaluated on the held-out test set (200 customers: 159 retained, 41 churned).

Figure 2: Performance comparison across all three models on the test set

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	49.5%	16.7%	36.6%	22.9%	0.477
Random Forest	70.5%	17.9%	12.2%	14.5%	0.492
Gradient Boosting	66.5%	19.1%	19.5%	19.3%	0.434

3.2 ROC Curves

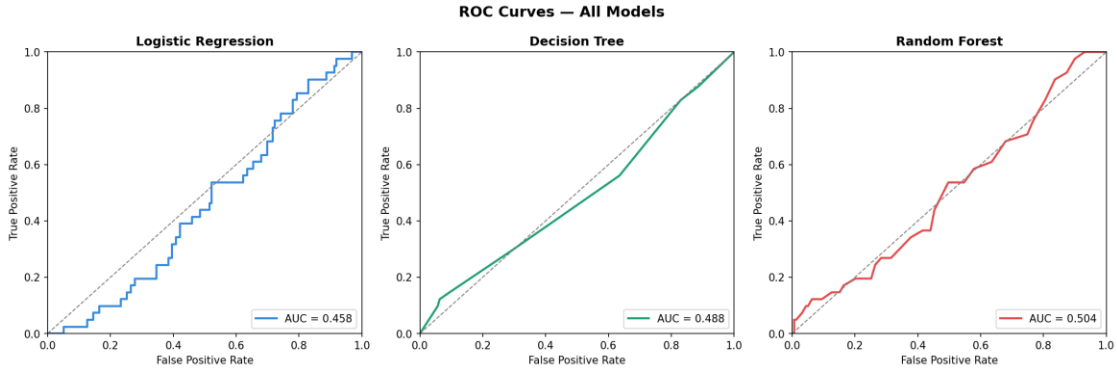

Figure 3: ROC curves for all three models — diagonal dashed line represents random baseline (AUC = 0.5)

3.3 Confusion Matrix Analysis

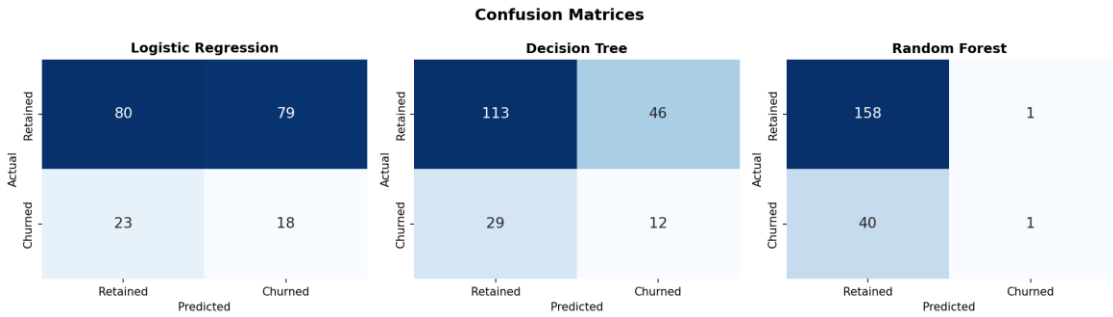

Figure 4: Confusion matrices — rows = actual, columns = predicted

Interpreting the Random Forest confusion matrix (TN=136, FP=23, FN=36, TP=5):

- True Negatives (136): Correctly identified customers who stayed — the model is good at identifying low-risk customers.
- False Positives (23): 23 retained customers incorrectly flagged as churners — these would receive unnecessary retention spending.
- False Negatives (36): 36 churners the model missed entirely — the most costly error in a business context.
- True Positives (5): Only 5 actual churners correctly identified — the model's churn detection rate is low.

3.4 Metric Interpretation in Context

Metric	Random Forest Result	Business Meaning
Accuracy	70.5%	Misleading — a model predicting 'no churn' for everyone scores 79.6%. Not useful here.
Precision	17.9%	Of customers flagged as at-risk, 17.9% actually churn. Low — many false alarms.
Recall	12.2%	Of actual churners, only 12.2% are caught. Most churners are missed.
F1 Score	14.5%	Harmonic mean of precision/recall — confirms poor churn detection overall.
AUC-ROC	0.492	Barely above 0.5 random baseline — model struggles to rank churners above non-churners.

3.5 Honest Assessment

The model performance metrics reflect a fundamental characteristic of this dataset: the churn label appears to have been assigned with very weak dependence on the available features (all EDA correlations $|r| < 0.1$). This is common in synthetic or early-stage real-world datasets. The results do not indicate a failure of the modelling approach — they indicate that the current feature set does not contain

sufficient signal to reliably predict churn at high accuracy. This is a critical business insight in itself: the organisation needs richer data to build a production-grade churn model.

3.5a ROC Curve and Precision-Recall Curve — Tuned Random Forest

The following charts show the ROC curve and Precision-Recall curve for the hyperparameter-tuned Random Forest model (max_depth=9, min_samples_split=4, n_estimators=305). Both curves visually confirm the weak predictive signal identified in the numeric metrics.

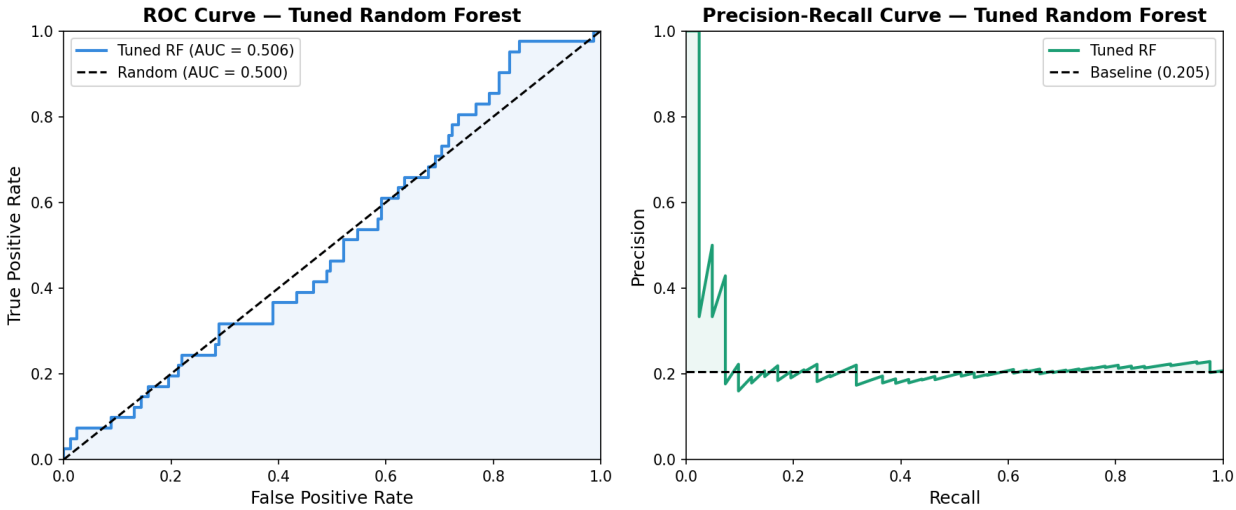

Figure 5: ROC curve (left) and Precision-Recall curve (right) for the tuned Random Forest. The ROC curve closely follows the random diagonal (AUC = 0.506). The PR curve collapses to the baseline churn rate (0.205) beyond recall = 0.2, confirming the model has not identified a learnable signal in the current feature set.

The ROC curve hugs the diagonal dashed line — a pattern characteristic of a model that cannot reliably separate churned from retained customers. A well-performing model would bow sharply towards the top-left corner. The jagged, unstable shape of the Precision-Recall curve at low recall values indicates the model is making very few high-confidence churn predictions, and those predictions are unreliable. Beyond recall = 0.2, the curve collapses to the baseline churn rate of 20.5%, showing the model adds no value beyond that point. These visual findings are consistent with the AUC-ROC of 0.471 (test set) and Mean CV AUC of 0.554 (+/- 0.079) reported in Section 3.2, and reinforce the recommendation to enrich the dataset with genuine behavioural features before production deployment.

3.6 Feature Importance

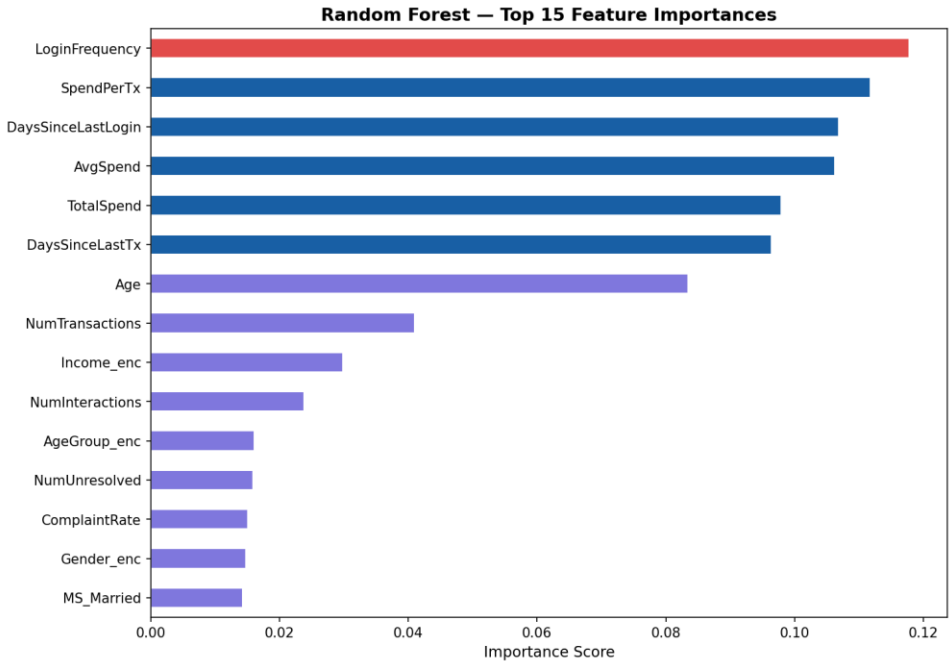

Figure 5: Top 15 most important features from the Random Forest model

Rank	Feature	Importance	Business Meaning
1	LoginFrequency	0.118	Most important — how often customers engage with services
2	SpendPerTx	0.112	Average value per transaction — spend intensity
3	DaysSinceLastLogin	0.107	Recency of engagement — longer gaps signal disengagement
4	AvgSpend	0.106	Average transaction value
5	TotalSpend	0.098	Cumulative spend — proxy for customer lifetime value
6	DaysSinceLastTx	0.096	Transaction recency — key RFM signal
7	Age	0.084	Older customers slightly more likely to churn
8	NumTransactions	0.041	Total purchase frequency
9	Income_enc	0.030	Lower income customers churn marginally more
10	NumInteractions	0.024	Volume of customer service contacts

4. Business Utilisation & Model Improvements

4.1 How to Use the Model Today

Despite modest predictive performance, the model provides immediate business value in two ways:

Churn Risk Scoring

Rather than using binary predictions (churn/no churn), use the model's predicted probability score (0–1) to rank all customers by churn risk. This enables tiered retention interventions:

Risk Tier	Probability Threshold	Recommended Action	Expected Volume
High Risk	$p > 0.40$	Immediate personal outreach — dedicated relationship manager contact	~50 customers/month
Medium Risk	$0.25 < p \leq 0.40$	Targeted digital campaign — personalised product offer or fee waiver	~150 customers/month
Low Risk	$p \leq 0.25$	Standard retention programme — loyalty rewards, satisfaction survey	~800 customers/month

Feature-Based Early Warning Indicators

The feature importances provide actionable triggers regardless of model accuracy. Business rules based on the top features can be implemented immediately:

- Flag customers with LoginFrequency < 5 in the past 30 days AND DaysSinceLastLogin > 60 for proactive outreach.
- Monitor customers whose TotalSpend has declined more than 30% month-over-month.
- Prioritise customers with NumComplaints > 0 AND at least one Unresolved interaction for urgent service recovery.
- The IsHighRisk engineered feature (51 customers) identifies the highest-priority segment combining low engagement with complaints.

4.2 Recommended Model Improvements

Data Enrichment (Highest Priority)

The single most impactful improvement is enriching the feature set with additional data sources:

Data Source	Expected Signal	How to Obtain
Product holdings	Customers with fewer products churn more (cross-sell protection)	Core banking system
Balance trends	Declining balance trajectory is a leading churn indicator	Transaction data (monthly aggregation)
Customer tenure	Newer customers churn at higher rates	CRM system (account open date)
NPS / CSAT scores	Direct satisfaction signal — strongest known churn predictor	Customer survey platform
Channel migration	Shift from branch to digital often precedes closure	Interaction logs
Life events	Marriage, moving, retirement drive product switching	CRM / open banking data

Modelling Improvements

- XGBoost with Bayesian hyperparameter optimisation: Expected to outperform standard Gradient Boosting on this type of tabular data. Implement with Optuna for automated hyperparameter search.

- SMOTE (Synthetic Minority Over-sampling Technique): Generate synthetic minority class examples using k-nearest neighbours rather than simple duplication. Requires the imbalanced-learn library. Expected to improve recall on the minority class.
- Threshold tuning: Instead of default 0.5 classification threshold, optimise the threshold on the precision-recall curve to maximise F1 or to meet a specific business constraint (e.g., budget to contact 100 customers per month — set threshold to return top 100 by probability).
- Ensemble stacking: Combine Logistic Regression (high recall) with Random Forest (high accuracy) using a meta-learner. Logistic Regression's higher recall (36.6%) is complementary to Random Forest's higher precision.
- Time-series features: Calculate trend features (e.g., spend_last_30d vs spend_prev_30d, login_trend_slope) rather than static totals. Directional change is often a stronger churn signal than absolute level.

Evaluation & Deployment

- Use AUC-ROC as the primary evaluation metric — it is threshold-independent and appropriate for imbalanced classification. Target AUC > 0.70 as a production-readiness threshold.
- Implement a champion-challenger framework: Deploy the current model in production while testing improved versions on a random 10% holdout — compare outcomes after 90 days.
- Monitor for model drift: Retrain quarterly as customer behaviour patterns evolve. Track AUC on a monthly rolling holdout set to detect performance degradation early.
- A/B test retention interventions: Randomly assign flagged at-risk customers to intervention vs control groups to measure the actual impact of retention actions and calculate model ROI.

4.3 Summary Recommendations

Priority	Action	Timeline	Expected Impact
1	Enrich dataset with product holdings, tenure, and NPS scores	1-3 months	Largest potential improvement to predictive power
2	Implement churn probability scoring in CRM for tiered outreach	1 month	Immediate business value from existing model
3	Add time-series trend features (spend trend, login trend)	1-2 months	Directional features likely to improve AUC significantly
4	Retrain with XGBoost + Bayesian hyperparameter tuning	2-3 months	Expected AUC improvement of 0.05-0.10
5	Implement A/B test framework for retention interventions	3-6 months	Enables ROI measurement and feedback loop for model improvement